

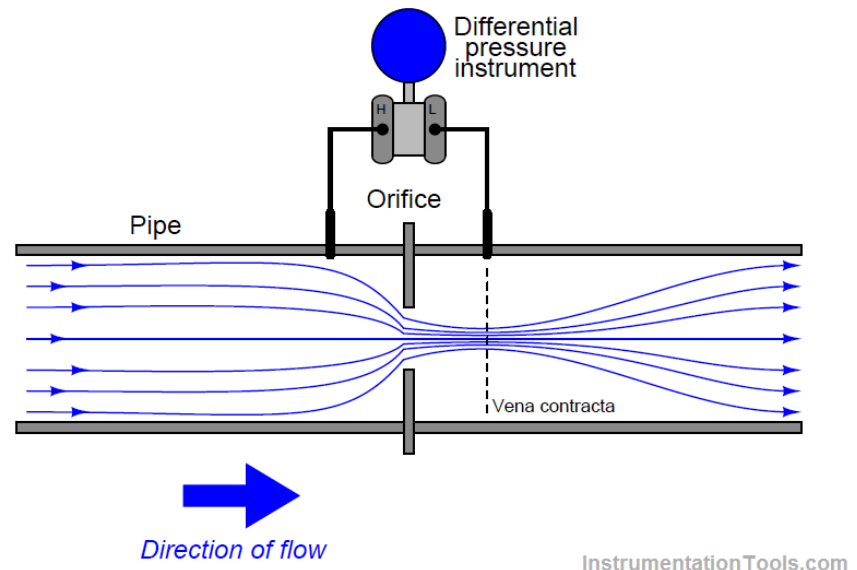
Errors during the measurement process

Sources of systematic error

□ System disturbance due to measurement



Heat transfer lowers the temperature of the water

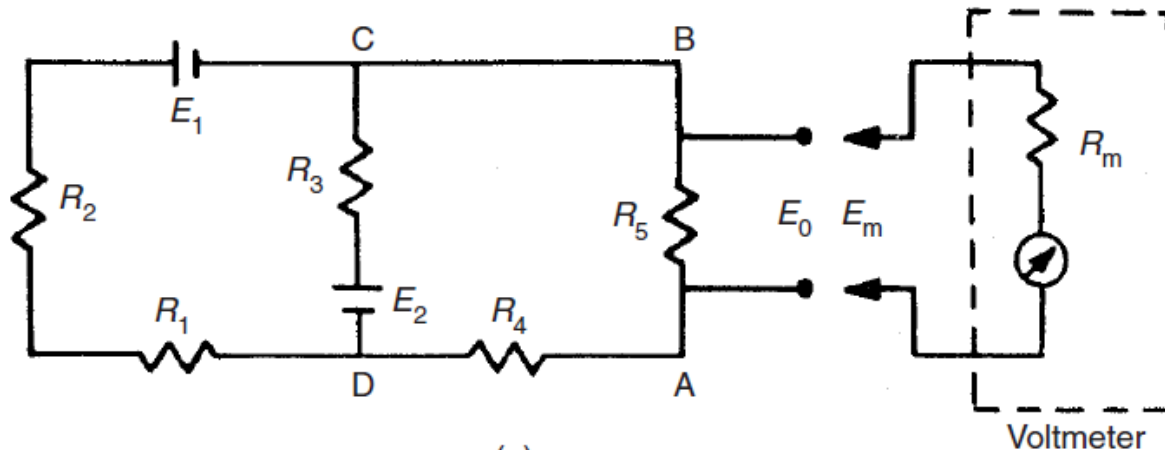


Permanent pressure loss in the flowing fluid

The process of measurement always disturbs the system being measured.

Sources of systematic error

□ Measurements in electric circuits

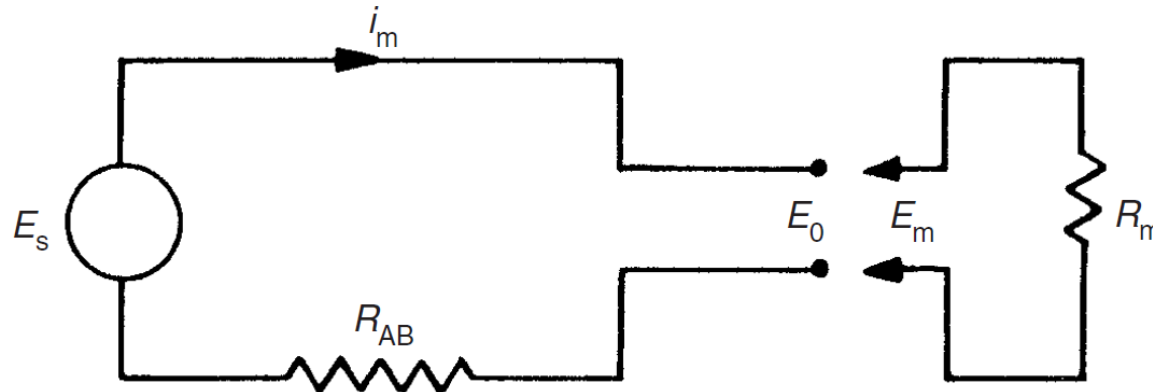


$$\frac{1}{R_{CD}} = \frac{1}{R_1 + R_2} + \frac{1}{R_3} \Leftrightarrow R_{CD} = \frac{(R_1 + R_2)R_3}{R_1 + R_2 + R_3}$$

$$\frac{1}{R_{AB}} = \frac{1}{R_{CD} + R_4} + \frac{1}{R_5} \Leftrightarrow R_{AB} = \frac{(R_4 + R_{CD})R_5}{R_4 + R_{CD} + R_5}$$

Sources of systematic error

□ Measurements in electric circuits

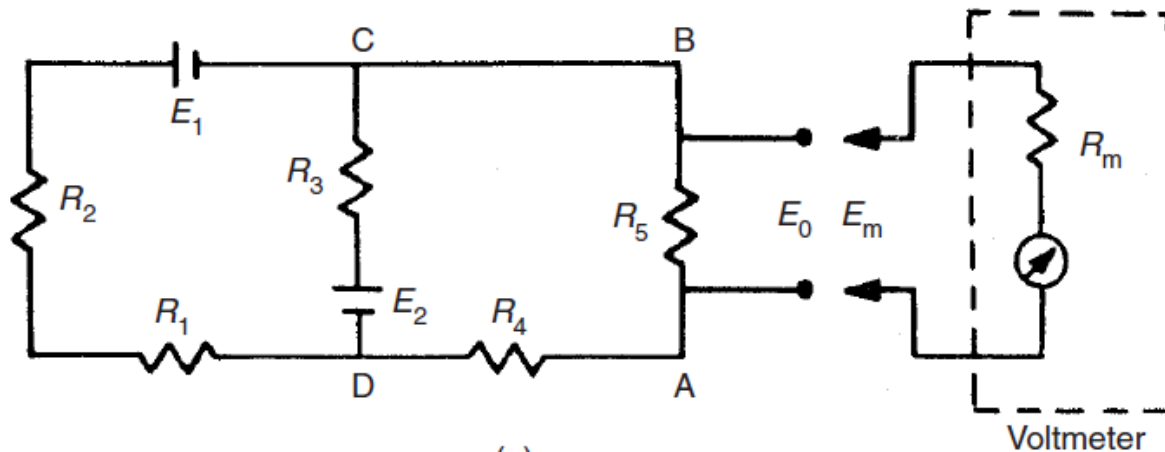


$$(b) \quad I = \frac{E_0}{R_{AB} + R_m}$$
$$E_m = \frac{R_m E_0}{R_{AB} + R_m} \Leftrightarrow \frac{E_m}{E_0} = \frac{R_m}{R_{AB} + R_m}$$

As R_m gets larger, the ratio E_m/E_0 gets closer to unity $\Rightarrow R_m$ should be as high as possible to minimize disturbance of the measured system.

Sources of systematic error

□ Measurements in electric circuits



Suppose that the components of the circuit the above figures have the following values:

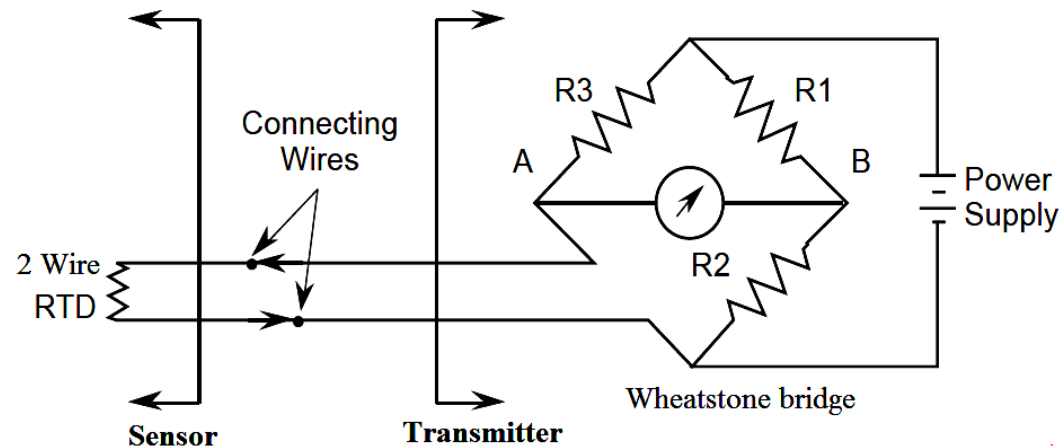
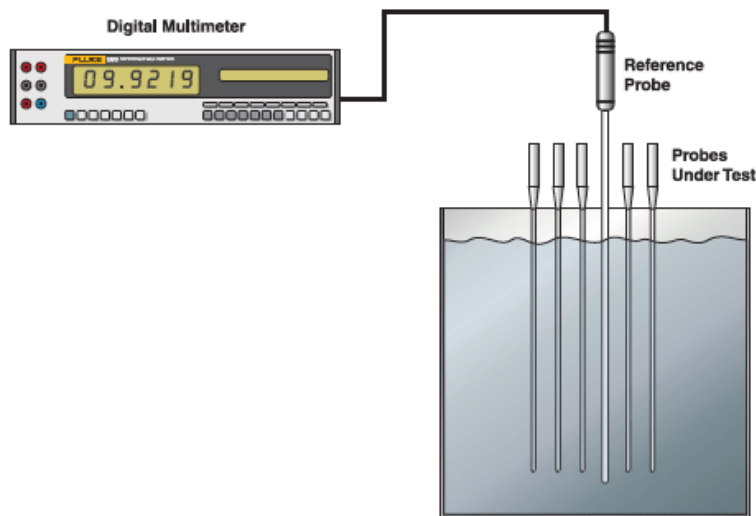
$$R_1 = 400\Omega; R_2 = 600\Omega; R_3 = 1000\Omega; R_4 = 500\Omega; R_5 = 1000\Omega$$

The voltage across AB is measured by a voltmeter whose internal resistance is 9500Ω .

What is the measurement error caused by the resistance of the measuring instrument?

Sources of systematic error

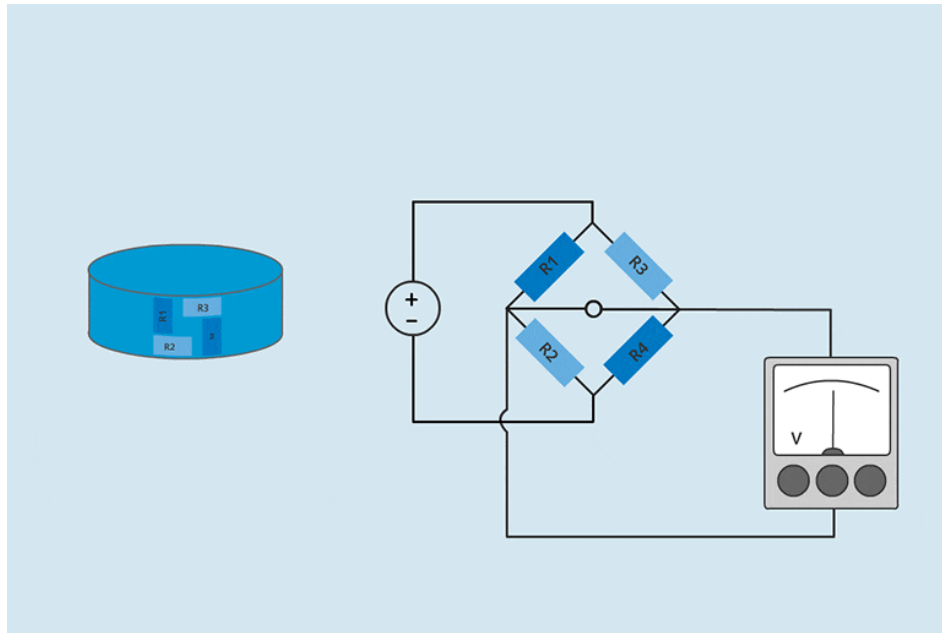
- ❑ Errors due to environmental inputs.
- ❑ Wear in instrument components
 - Connecting leads: the resistance of connecting leads (or pipes in the case of pneumatically or hydraulically actuated measurement systems).



Reduction of systematic errors

□ Careful instrument design

- Reducing the sensitivity of an instrument to environmental inputs to as low a level as possible.

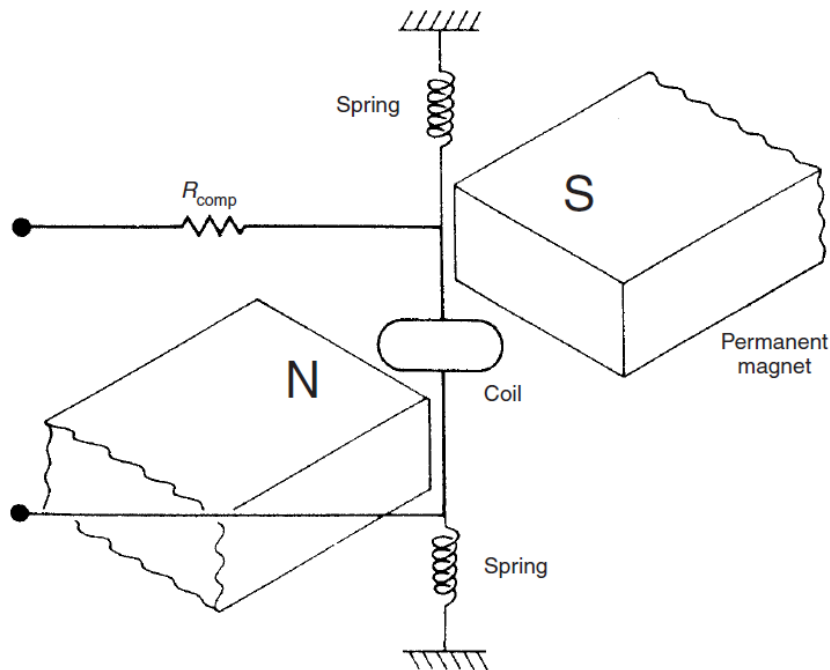


- In the design of strain gauges, the element should be constructed from a material whose resistance has a very low temperature coefficient (i.e. the variation of the resistance with temperature is very small).

Reduction of systematic errors

□ Method of opposing inputs

- The method of opposing inputs compensates for the effect of an environmental input in a measurement system by introducing an equal and opposite environmental input that cancels it out.



- In response to an increase in temperature,

$$R_{coil} \uparrow$$
$$R_{comp} \downarrow$$

- The total resistance remains approximately the same

Fig. 3.2 Millivoltmeter.

Reduction of systematic errors

□ Example of 3-wire bridge circuit

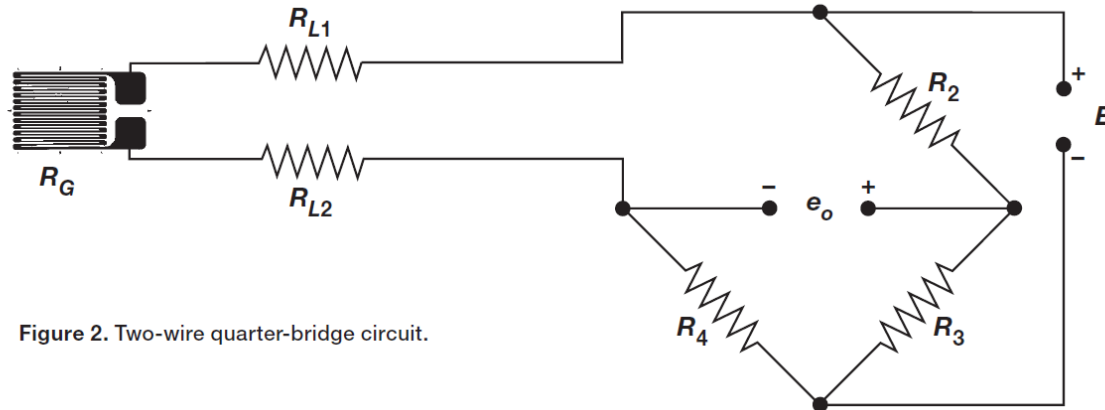


Figure 2. Two-wire quarter-bridge circuit.

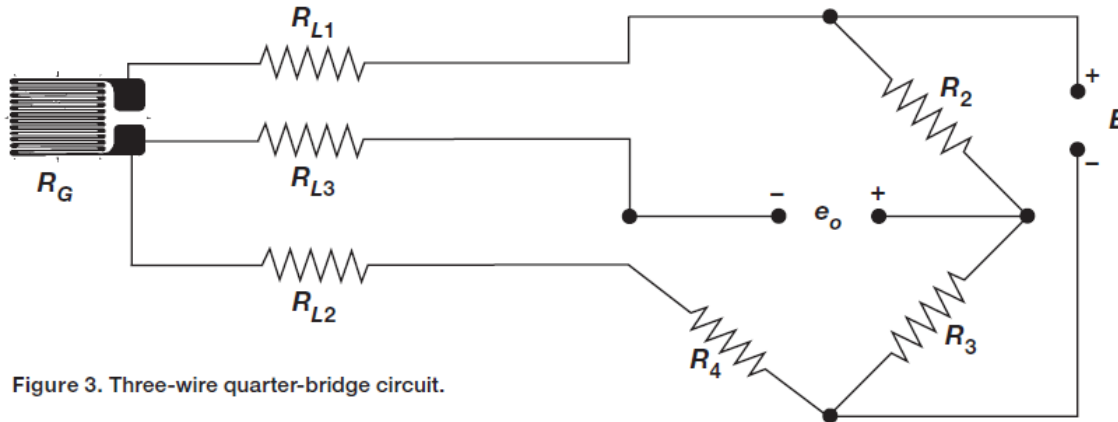
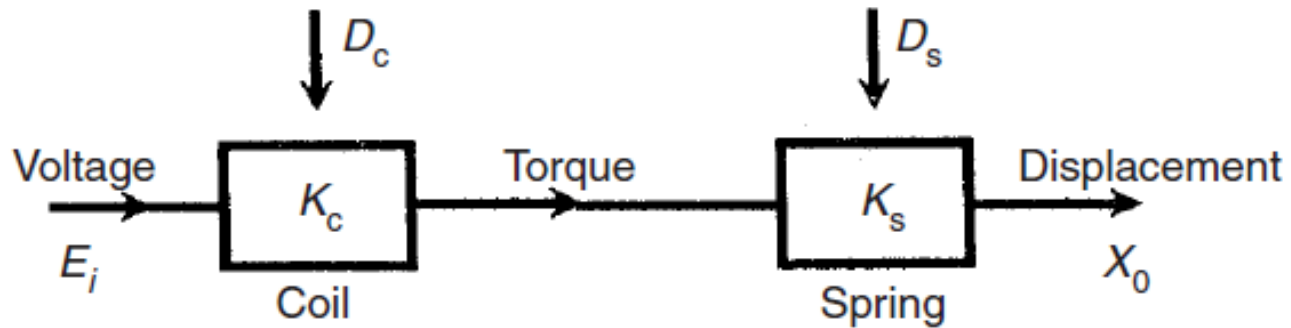


Figure 3. Three-wire quarter-bridge circuit.

Reduction of systematic errors

□ High-gain feedback



Reduction of systematic errors

□ High-gain feedback

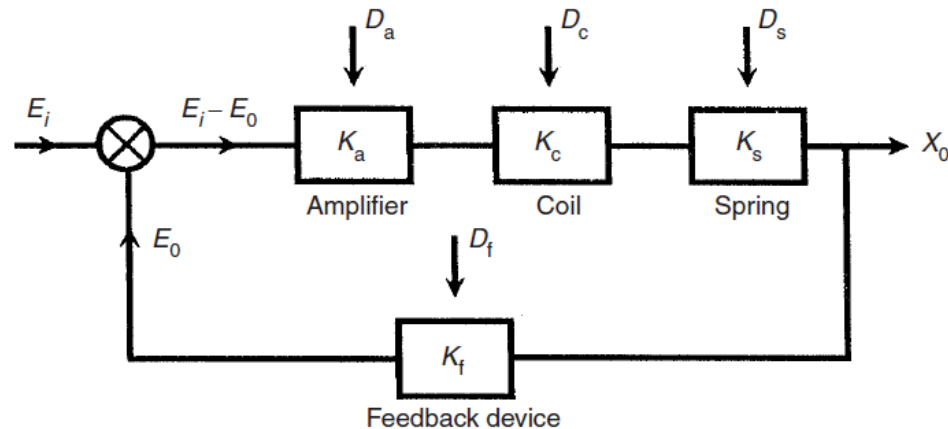


Fig. 3.4 Block diagram of voltage-measuring instrument with high-gain feedback.

$$E_0 = K_f X_0; X_0 = (E_i - E_0) K_a K_m K_s = (E_i - K_f X_0) K_a K_m K_s$$

Thus:

$$E_i K_a K_m K_s = (1 + K_f K_a K_m K_s) X_0$$

$$X_0 = \frac{K_a K_m K_s}{1 + K_f K_a K_m K_s} E_i$$

Because K_a is very large, $K_f K_a K_m K_s \gg 1$

$$X_0 = E_i / K_f$$

Random Error

□ Mean

$$x_{mean} = \frac{x_1 + x_2 + \dots x_n}{n}$$

□ Standard deviation and variance

$$d_i = x_i - x_{mean}$$

Variance

$$V = \frac{d_1^2 + d_2^2 \dots d_n^2}{n - 1}$$

Standard deviation:

$$\sigma = \sqrt{V} = \sqrt{\frac{d_1^2 + d_2^2 \dots d_n^2}{n - 1}}$$

Aggregation of measurement system errors

□ Combined effect of systematic and random errors

■ Systematic errors: $\pm x$

■ Random errors: $\pm y$

Likely maximum error:

$$e = \sqrt{x^2 + y^2}$$

Aggregation of measurement system errors

□ Error in a sum

Two outputs y and z of separate measurement system components are to be added together $S = y + z$

If the maximum errors in y and z are $\pm ay$ and $\pm bz$:

$$S_{max} = (y + ay) + (z + bz)$$

$$S_{min} = (y - ay) + (z - bz)$$

$$S = y + z \pm (ay + bz)$$

Absolute errors

$$e = \sqrt{(ay)^2 + (bz)^2}$$

Thus $S = (y + z) \pm e$. This can be express in alternative form:

$$S = (y + z)(1 \pm f) \text{ where } f = e/(y + z)$$

Aggregation of measurement system errors

□ Error in a sum

Example:

A circuit requirement for a resistance of 550Ω is satisfied by connecting together two resistors of nominal values 220Ω and 330Ω in series. If each resistor has a tolerance of $\pm 2\%$, find the error in the sum and the total resistance.

The error in the sum:

$$e = \sqrt{(0.02 \times 220)^2 + (0.02 \times 330)^2} = 7.93; f = \frac{7.93}{550} = 0.0144$$

The total resistance S can be expressed as

$$S = 550\Omega \pm 7.93\Omega \text{ or } S = 550(1 \pm 0.0144)\Omega \text{ i.e. } S = 550\Omega \pm 1.4\%$$

Aggregation of measurement system errors

□ Error in a difference

Two outputs y and z of separate measurement system components are to be subtracted, $S = y - z$

If the maximum errors in y and z are $\pm ay$ and $\pm bz$:

$$S = (y - z) \pm e \text{ or } S = (y - z)(1 \pm f)$$

Where e is $e = \sqrt{(ay)^2 + (bz)^2}$, and $f = e/(y - z)$

Aggregation of measurement system errors

□ Error in a difference

Example

A fluid flow rate is calculated from the difference in pressure measured on both sides of an orifice plate. If the pressure measurements are 10.0 bar and 9.5 bar and the error in the pressure measuring instruments is specified as $\pm 0.1\%$, find e and f ?

$$e = \sqrt{(0.001 \times 10)^2 + (0.001 \times 9.5)^2} = 0.0138$$
$$f = 0.0138/0.5$$

Aggregation of measurement system errors

□ Error in a product

If the outputs y and z of two measurement system components are multiplied together, $P=yz$.

If the possible error in y is $\pm ay$ and in z is $\pm bz$

$$P_{max} = (y + ay)(z + bz) = yz + ayz + byz + aybz$$

$$P_{min} = (y - ay)(z - bz) = yz - ayz - byz + aybz$$

For $a \ll 1, b \ll 1$

$$P_{max} = yz(1 + a + b); P_{min} = yz(1 - a - b)$$

Likely maximum error in P :

$$e = \sqrt{a^2 + b^2}$$

Aggregation of measurement system errors

□ Error in a product

Example

If the power in a circuit is calculated from measurements of voltage and current in which the calculated maximum errors are respectively $\pm 1\%$ and $\pm 2\%$, find the maximum likely error?

$$e = \pm \sqrt{0.01^2 + 0.02^2} = \pm 0.022 \text{ or } \pm 2.2\%$$

Aggregation of measurement system errors

□ Error in a quotient

If the output measurement y of one system component with possible error $\pm x$ is divided by the output measurement z of another system component with possible error $\pm bz$.

The maximum and minimum possible values for the quotient can:

$$Q_{max} = \frac{y + ay}{z - bz} = \frac{(y + ay)(z + bz)}{(z - bz)(z + bz)} = \frac{yz + ayz + byz + aybz}{z^2 - b^2z^2}$$
$$Q_{min} = \frac{y - ay}{z + bz} = \frac{(y - ay)(z - bz)}{(z + bz)(z - bz)} = \frac{yz - ayz - byz + aybz}{z^2 - b^2z^2}$$

For $a \ll 1$ and $b \ll 1$

$$Q_{max} = \frac{yz(1+a+b)}{z^2}; \quad Q_{min} = \frac{yz(1-a-b)}{z^2}; \quad Q = \frac{y}{z} \pm \frac{y}{z}(a + b)$$

Aggregation of measurement system errors

□ Error in a quotient

Example

If the density of a substance is calculated from measurements of its mass and volume where the respective errors are $\pm 2\%$ and $\pm 3\%$. Find the maximum likely error in the density value?

$$e = \pm \sqrt{0.02^2 + 0.003^2} = \pm 0.036 \text{ or } \pm 3.6\%$$

Total error when combining multiple measurements

Example

A rectangular-sided block has edges of lengths a , b and c , and its mass is m . If the values and possible errors in quantities a , b , c and m are as shown below, calculate the value of density and the possible error in this value.

$$a = 100\text{mm} \pm 1\%; b = 200\text{mm} \pm 1\%; c = 300\text{mm} \pm 1\%; m = 20\text{kg} \pm 0.5\%$$

- $ab = 0.02 \text{ m}^2 \pm 2\%$ (possible error = $1\%+1\%=2\%$)
- $(ab)c = 0.006 \text{ m}^3 \pm 3\%$ (possible error = $2\% + 1\% = 3\%$)
- $\frac{m}{abc} = \frac{20}{0.006} = \frac{3330\text{kg}}{\text{m}^3} \pm 3.5\%$ (possible error = $3\% + 0.5\%=3.5\%$)